THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 9: **Propositions and their Converses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If n is any integer not divisible by 3, then n^2 leaves a remainder ___ on division by 3. For 14 this gives: 1

(a) 0 **(b)** 3 **(c)** 2 **(d)** 1

2. For the two-digit number 94, the value of (number − digit sum) ÷ 9 is: 1

(a) 9 **(b)** 10 **(c)** 8 **(d)** 11

3. If the sides of a triangle are 36, 77 and x (with x the longest side) and it is right-angled, then x is: 1

(a) 87 **(b)** 84 **(c)** 86 **(d)** 85

4. The smallest natural number with exactly 10 factors (counting 1 and itself) is: 1

(a) 48 **(b)** 52 **(c)** 49 **(d)** 46

5. The number of perfect squares between 900 and 5000 (both excluded) is: 1

(a) 41 **(b)** 42 **(c)** 39 **(d)** 40

6. For the expression $n^2 - n + 137$, the smallest natural number n for which the value is NOT prime is: 1

(a) 2 **(b)** 4 **(c)** 3 **(d)** 5

7. For the expression $n^2 - n + 197$, the smallest natural number n for which the value is NOT prime is: 1

(a) 4 **(b)** 5 **(c)** 3 **(d)** 2

8. The sum of 11 consecutive integers is 682. The largest of them is: 1

(a) 69 **(b)** 67 **(c)** 66 **(d)** 68

9. **Assertion (A):** The statement '$n^2 > n$ for every natural number n' is false. 1

Reason (R): For $n = 1$ we get $1^2 = 1$, which is not greater than 1.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The converse of 'If a quadrilateral is a rectangle, then its opposite sides are equal' is false. 1
Reason (R): The converse of 'If P, then Q' is 'If Q, then P'.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Verify that $n^2 - 1$ is a multiple of 8 for the odd number $n = 19$. Write the converse of "If n is odd, then $n^2 - 1$ is a multiple of 8". 2

12. Use the converse of the Baudhāyana–Pythagoras theorem: for what value of x is a triangle with sides $33, 56, x$ (x the longest) right-angled? 2

13. Show that "If a number is divisible by 6 and by 8, then it is divisible by 48" is false. Give a counterexample. 2

14. Verify that $n^2 - 1$ is a multiple of 8 for the odd number $n = 25$. Write the converse of "If n is odd, then $n^2 - 1$ is a multiple of 8". 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that the sum of the squares of two consecutive integers is odd. Verify it for 24 and 25. 3

16. Prove: if 4 divides x and x divides y, then 4 divides y. Verify with $x = 20$ and $y = 180$. 3

17. For $m = 9$ and $n = 2$ show that $(77,\ 36,\ 85)$ are the sides of a right triangle, using the converse of the Baudhāyana–Pythagoras theorem. 3

OR

Prove that the sum of the squares of two consecutive integers is odd. Verify it for 31 and 32. 3

18. Prove: if 6 divides x and x divides y, then 6 divides y. Verify with $x = 18$ and $y = 144$. 3

OR

For $m = 10$ and $n = 7$ show that $(51,\ 140,\ 149)$ are the sides of a right triangle, using the converse of the Baudhāyana–Pythagoras theorem. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Karan is studying quadrilaterals and writes: 'If a quadrilateral is a square, then all its sides are equal.' 5
(i) Write the converse.
(ii) Is the original statement true?
(iii) Is the converse true?
(iv) Give a counterexample to the converse and explain why it works.

20. Prove that the product of three consecutive integers is divisible by 6. Verify it for $28, 29, 30$. 5

OR

Prove that the square of any odd integer is 1 more than a multiple of 8. Verify it for 33. 5

END OF QUESTION PAPER — Total: 40 Marks